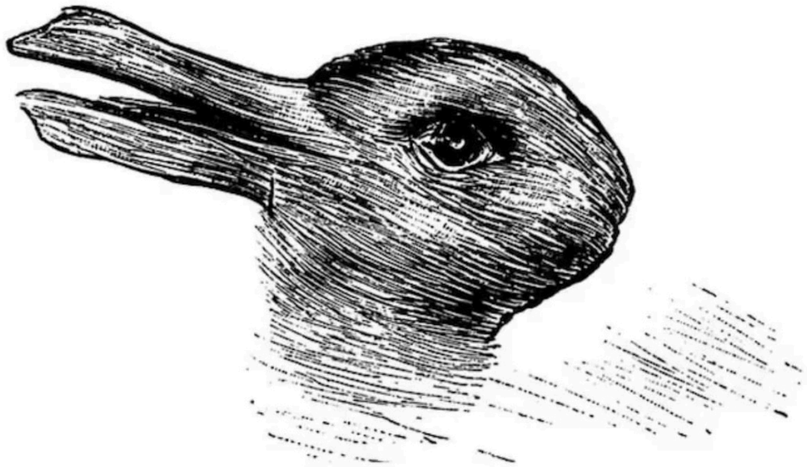
Redundancy of length in physics



Tsampodimos, Apostolos – apostolos.tsampodimos@ftmc.lt

Supervisor: Karčiauskas, Mindaugas

Why modified gravity?

General relativity

classical theory

geometrical description of gravity

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Why modified gravity?

General relativity

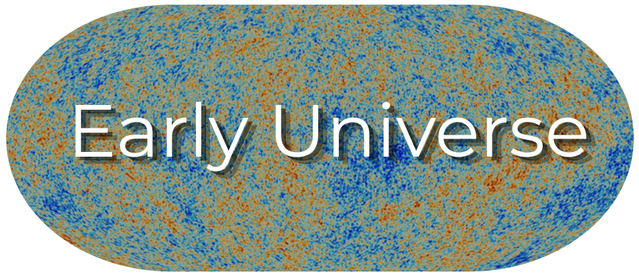
classical theory

geometrical description of gravity

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classical theory

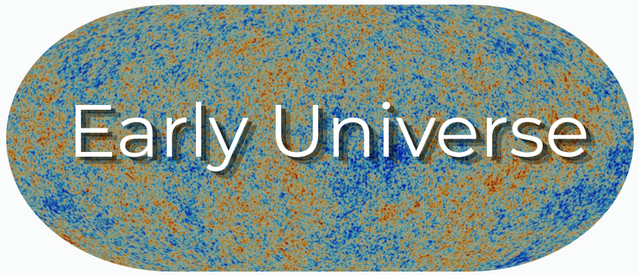
geometrical description of gravity

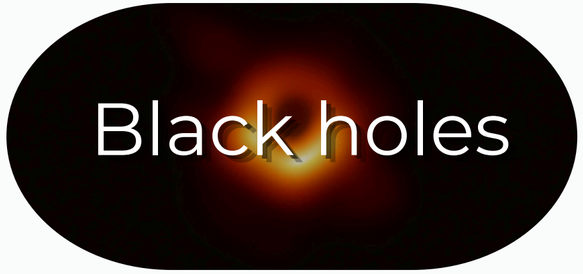
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Quantum mechanics

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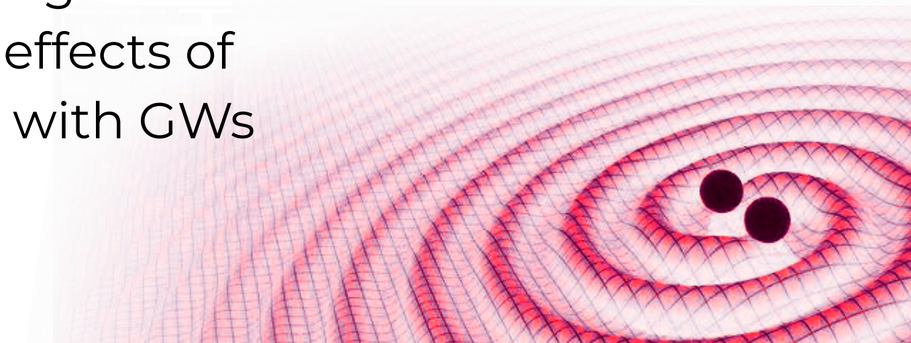
Dark energy

Early Universe

Black holes

Quantum mechanics

Probing non-linear effects of gravity with GWs



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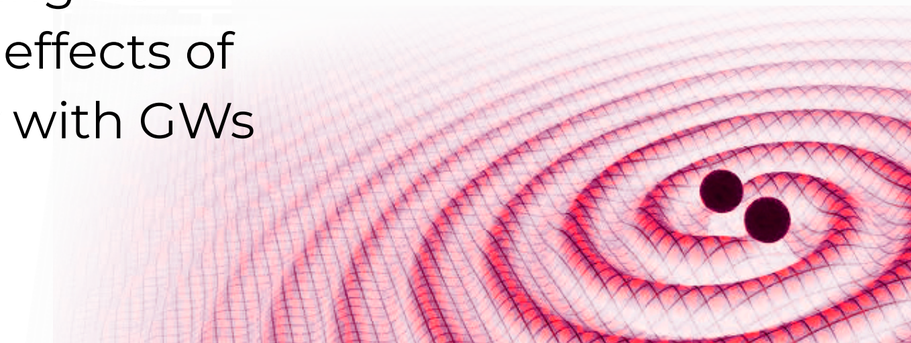
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Motivation for **modified gravity**

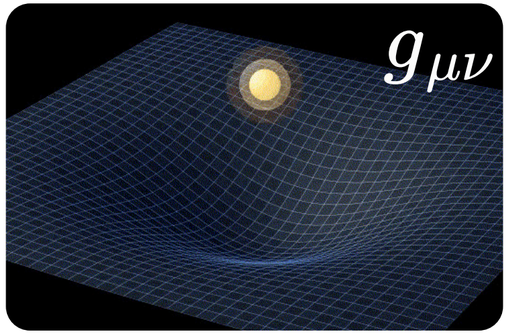


Geometry

metric

↓

geometry of spacetime

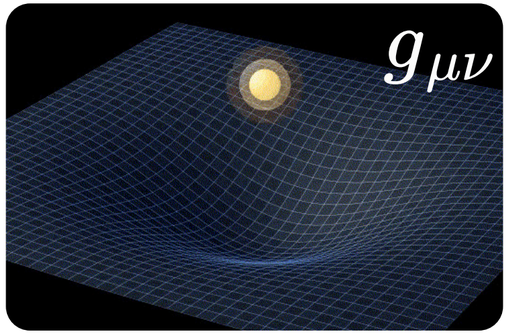


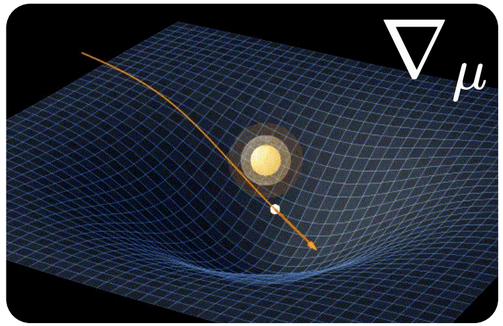
Geometry

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geometry of spacetime





connection

↓

how to move in spacetime

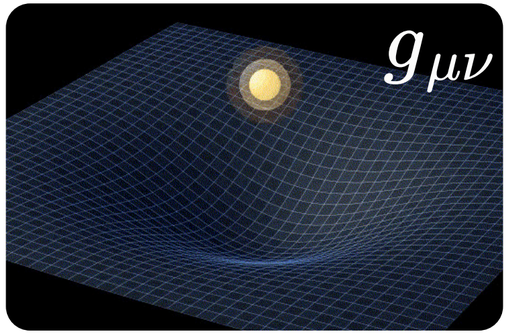
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$$\mathrm{G}[g_{\mu\nu}] = \mathrm{T}_{\mu\nu}$$

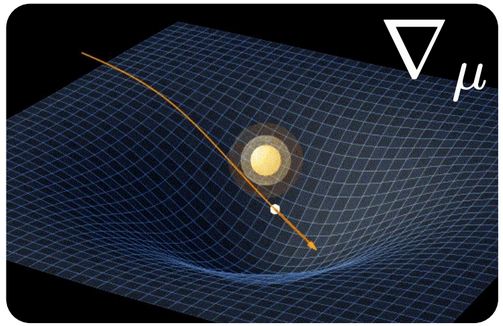
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geometry of spacetime



$$(\mathcal{M}, \nabla, g)$$



connection

↓

how to move in spacetime

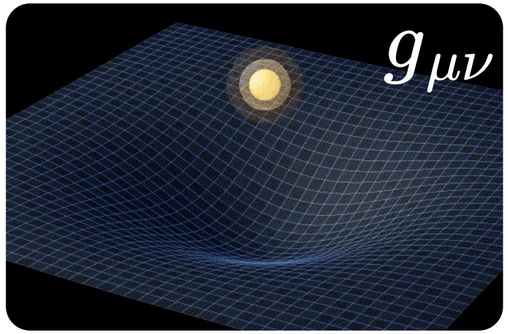
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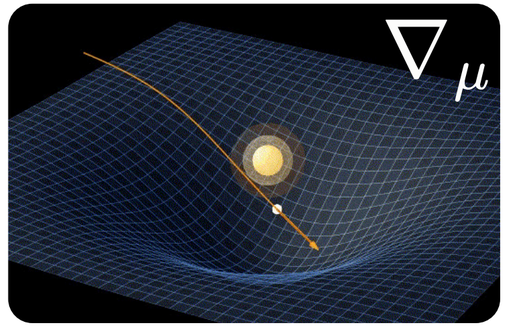
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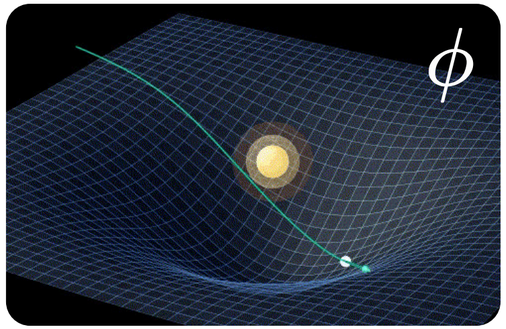


connection

↓

how to move in spacetime

↓ + scale



Integrable Weyl geometry

$$\mathrm{G}[g_{\mu\nu}] - 2(\nabla_\mu \nabla_\nu \phi - \partial_\mu \phi\, \partial_\nu \phi) + (2\, g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi + g^{\alpha\beta} \partial_\alpha \phi\, \partial_\beta \phi) g_{\mu\nu} = e^{-2\phi}\, \mathrm{T}_{\mu\nu}$$

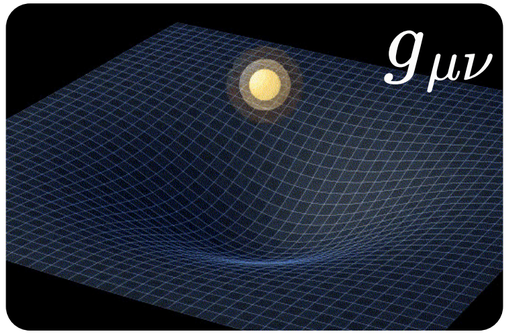
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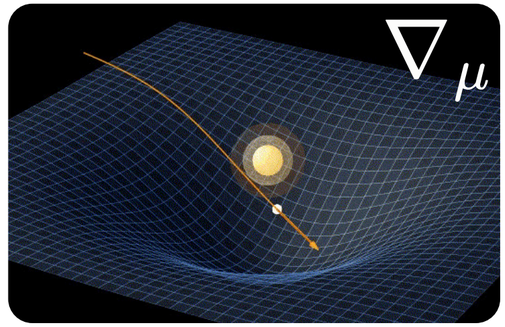
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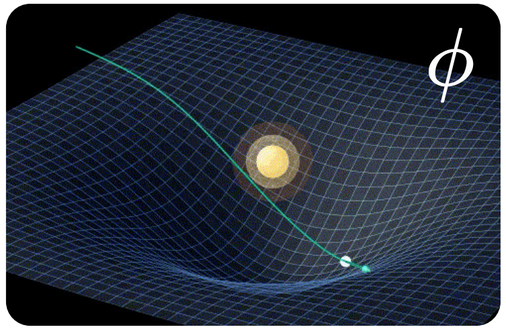


connection

↓

how to move in spacetime

↓ + scale



Integrable Weyl geometry

$$(\mathcal{M}, \nabla, g, \phi)$$

We make the choice of scale explicit

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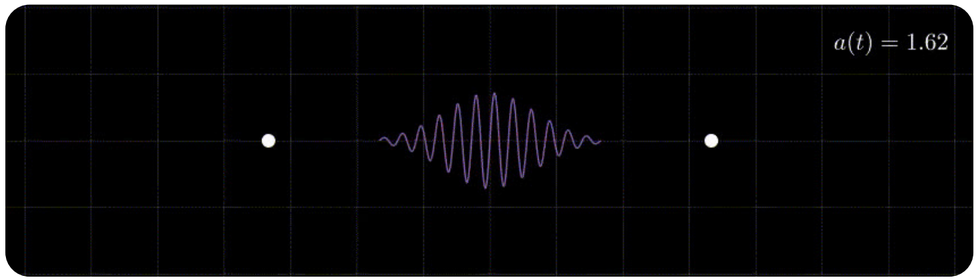
Interpretation

$$1 + z = \frac{\nu}{\nu_0} = \frac{e^{-\phi}\,\omega[g_{\mu\nu}]\big|_e}{e^{-\phi}\,\omega[g_{\mu\nu}]\big|_o}$$

redshift

Interpretation

$$(g^{\mathrm{FLRW}}_{\mu\nu}, 0)$$



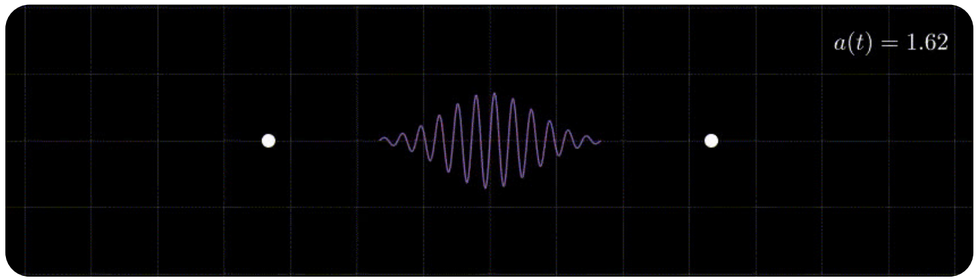
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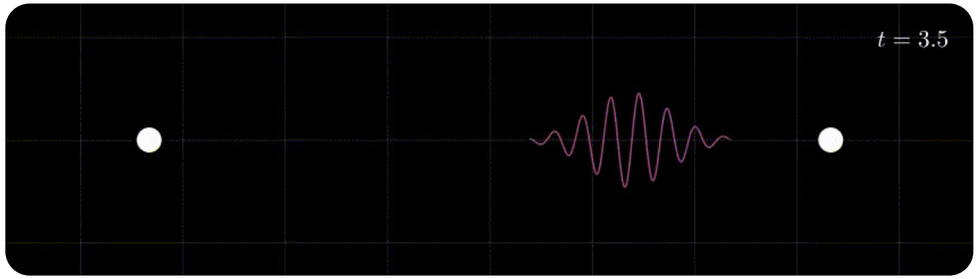
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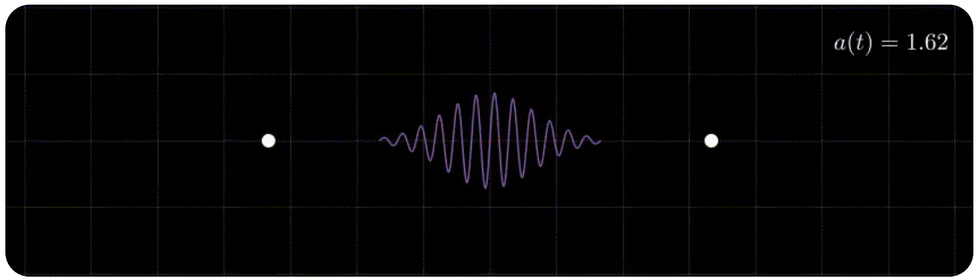


time dependent electron mass

$$(\eta_{\mu\nu}, ln(a))$$

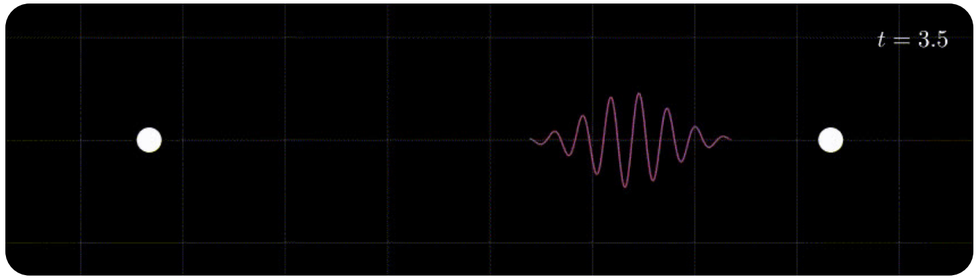
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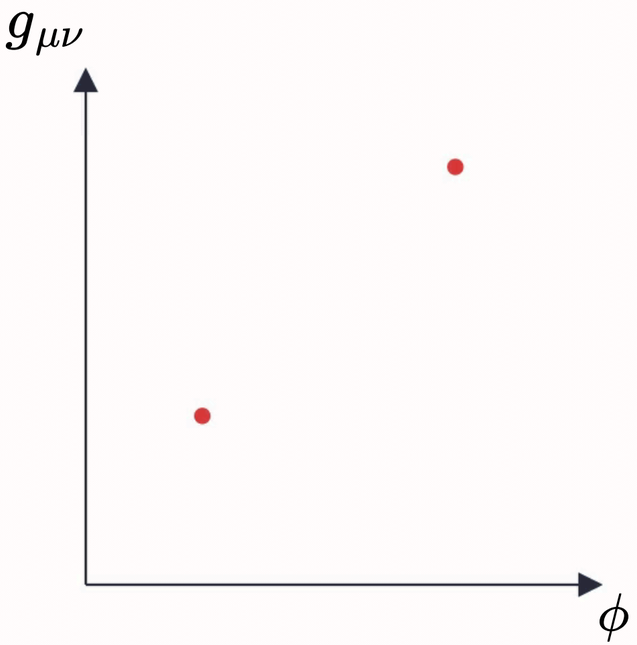
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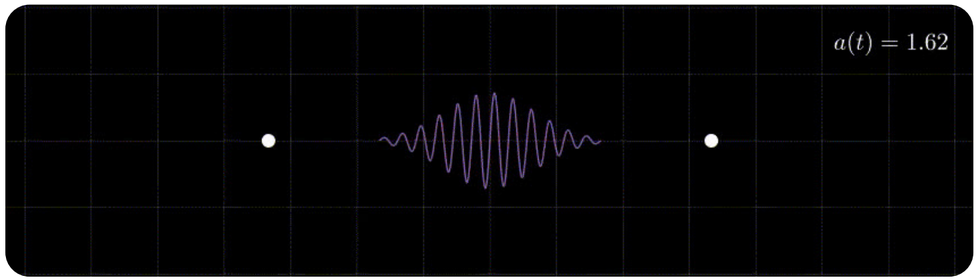
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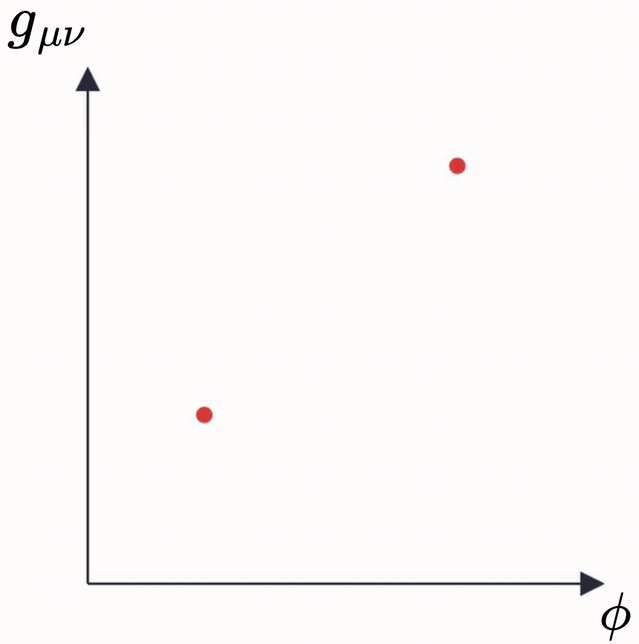


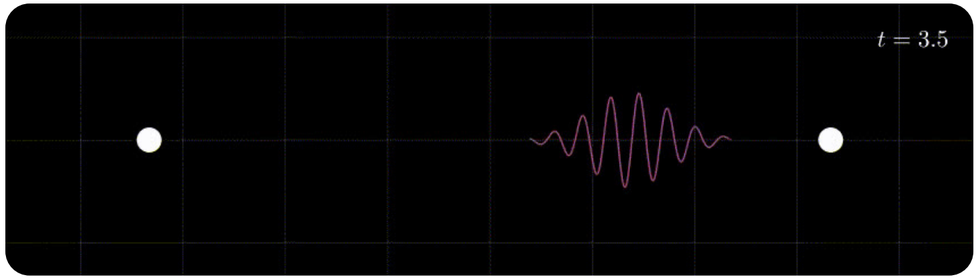
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Interpretation = choosing different pairs of (metric, scale)

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redshift





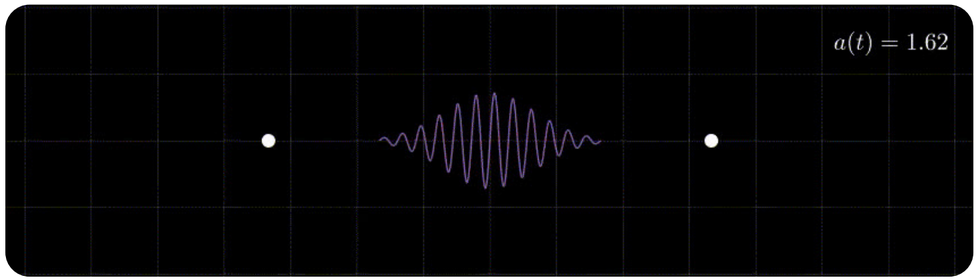
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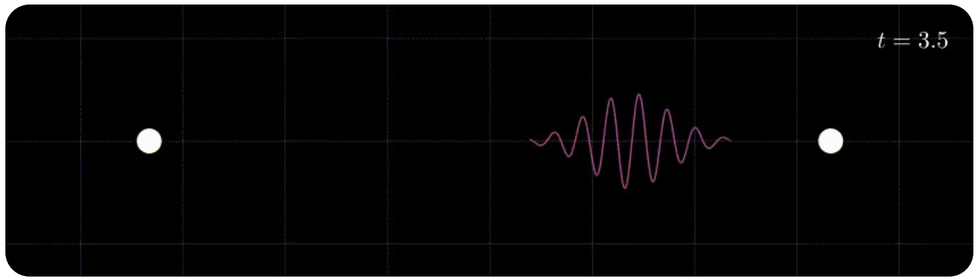
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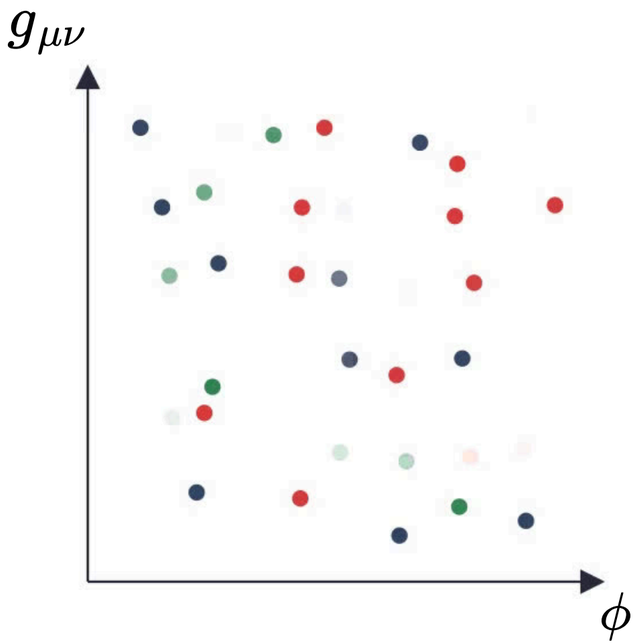
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Theories



Observables

PhysRevD.65.124013

Partial: A physical quantity to which we associate a (measuring) procedure.

mass

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redshift

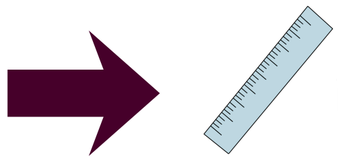
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independent = “interpretation invariant”

$$\iota : \mathcal{A}_{\mathrm{D}}^{0} \mapsto \mathcal{A}_{\mathrm{D}}^{\mathrm{ext}}$$

Observables

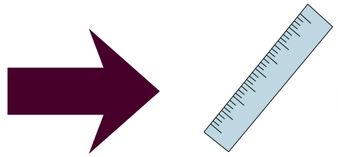
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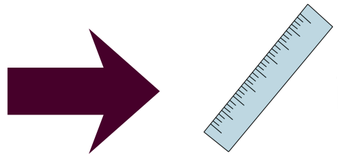
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$$\mathrm{H}_T \equiv N\left(-\frac{p_a^2}{12M_{\mathrm{Pl}}^2 a} + \frac{p_\chi^2}{2M_{\mathrm{Pl}}^2 a^3} + M_{\mathrm{Pl}}^4 a^3 V(\chi) + M_{\mathrm{Pl}} n_0 \mu(\chi)\right) + \lambda(a p_a - p_\sigma)$$

$\mathbb{R} \times \mathbb{T}^3$

Wrap up

Outlook

1. What happens when going quantum?

2. Generalizing to the lightcone.

Wrap up

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Conclusions

1. Scale invariance of complete observables.
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Thank you